

IEEE UV2026



General Information



IEEE UV2026 Organizing Committee

IEEE UV Conference Board

<https://universalvillage.org/ieee-uv-conference>

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THE 8TH
IEEE UV2026
INTERNATIONAL CONFERENCE ON UNIVERSAL VILLAGES



OCTOBER 17-20 | BOSTON AND VIRTUAL



UV Concept & IEEE UV Conferences



- ❖ **Universal Village (UV)** is a new concept proposed by **MIT's Universal Village Program** in 2012. UV exemplifies an ideal future society that pursues harmony between humans and nature through the wise use of advanced technologies.
- ❖ Led by the **MIT** professors who initiated the **MIT UV** program as well as the UV conferences, the **IEEE UV Conferences** call for collective efforts from world-renowned experts across multi-disciplinary fields to collaborate beyond traditional field boundaries. Previous IEEE UV conferences (**UV2013, UV2014, UV2016, UV2018, UV2020, UV2022** and **UV2024**) were groundbreaking international venues that inspired successful collaboration among governments, academia, industries, and civil society.

<https://universalvillage.org/ieee-uv-conference>

Universal Village (UV) & Future Lifestyles



Universal

- ❖ Following the laws of the universe

Village

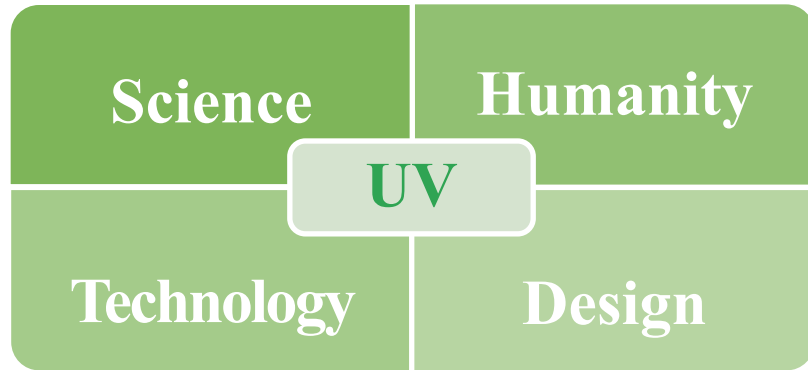
- ❖ Cities are not the only living environments



**Universal
Village**

- ❖ New lifestyles beyond Smart Cities following the laws of the universe

UV Elements



4 Impacting Factors

- ❖ UV Information Flow
- ❖ UV Material Cycles
- ❖ UV Lifestyles
- ❖ UV Communities

8 UV Sub-Systems

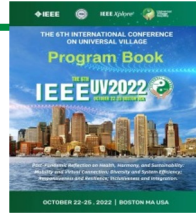
1. Smart Home and Community
2. Smart Medicine and Healthcare
3. ITS, Urban Planning and Crowd Management
4. Smart Energy Management
5. Smart City Infrastructure
6. Smart Response System for City Emergencies
7. Smart Environmental Protection
8. Smart Humanity

UV Scope



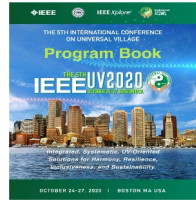
Previous UV Conferences (UV2013-UV2022)

UV2022
Virtual



POST-PANDEMIC REFLECTION ON HEALTH, HARMONY, AND SUSTAINABILITY: MOBILITY & VIRTUAL CONNECTION; DIVERSITY & SYSTEM EFFICIENCY; RESPONSIVENESS & RESILIENCE; INCLUSIVENESS & INTEGRATION

UV2020
Boston



INTEGRATED, SYSTEMATIC, UV-ORIENTED, SOLUTIONS FOR HARMONY, RESILIENCE, INCLUSIVENESS AND SUSTAINABILITY

UV2018
Boston



DIVERSITY, MOBILITY AND CONNECTIVITY FOR EMERGENCE OF HARMONY IN UNIVERSAL VILLAGE

UV2016
Nagoya



UV FOR REGIONS AT DIFFERENT DEVELOPMENT PHASES, PEOPLE AT DIFFERENT AGES WITH VARIOUS TECHNOLOGY & CULTURAL BACKGROUNDS

UV2014
Boston



INNOVATE FUTURE LIFESTYLE FOR SUSTAINABLE HAPPINESS

UV2013
Beijing



HUMANKIND IN HARMONY WITH NATURE THROUGH WISE USE OF TECHNOLOGY

Previous UV Conferences (UV2024)

Reflection of Human - AI Relationship and AI-Pervasive Society Transformation:

*Human-centricity, Explainable AI,
and Intent & Behavior Understanding
for Integration, Resilience, Inclusiveness, and Sustainability*

IEEE UV2026 Themes



Human-Centered AI Transformation for Future Empowerment:

- ❖ *Explainable & Human-Intelligible Understanding*
- ❖ *Intent-Aware Reasoning & Accountable Decision-Making*
- ❖ *Human-Authorized AI Action*
- ❖ *Human-Supervised Reflective Learning*

Introduction to IEEE UV2026



After the great success of **IEEE UV2024** and **IEEE UV2022**, the **8th International Conference on Universal Village (IEEE UV2026)** will be held both **Virtually** and **Locally** from October 17-20, 2026.

Subjects for IEEE UV2026 include but are not limited to the following topics:

- ❖ The Human - AI Relationship and AI-Pervasive Societal Transformation
- ❖ Developing systematic methodologies to advance UV technologies and to develop UV systems, including Intelligent Transportation Systems, Urban Planning and Crowd Management, Smart Infrastructure, Smart Response Systems for Emergencies, Smart Energy Management, Smart Environmental Protection, Smart Home and Community, Smart Medicine and Healthcare, and Smart Humanity
- ❖ New lifestyles enabled by IT / New energy sources / New materials / Effective microorganism technology and environmental protection
- ❖ Exploring UV development paths that are appropriate for different regions at different development phases, from theoretical exploration, research development, framework design, implementation, engineering, to integration.
- ❖ Inspiring successful collaboration among governments, academia, industries, and civil society to address different UV challenges and seek practical UV solutions

UV2026 Program Committees (TBD)

2026
CONFERENCE ON UNIVERSAL VILLAGE
VIRTUAL AND LOCAL



Program Chairs

Shengsheng Cao (Ningbo Univ., China)

Zili Li (MIT, USA)

Hao Sheng (Beihang Univ., China)

Hiro Aoki (Nagoya Univ., Japan)

Wenya Du (UVS, USA)

Yi Tao (UVS, USA)

Juntao Jiang (Zhejiang Univ., China)

Zhenyao Liu (UPenn, USA)

Publication Chairs

HaoYuan (Stephen Inst. of Tech., USA)

Hongyan Cui (MIT&BUPT, USA)

Lifeng Zhang (BJMU, China)

IT Chairs

Jieren Kou (NYU, USA)

Hanxia Li (UVS, USA)

Zhenqian Huang (Hohai Univ., China)

Chuoqiao Gu (Google, USA)

UV2026 Executive Committees (TBD)

2026
CONFERENCE ON UNIVERSAL VILLAGE
VIRTUAL AND LOCAL



Honorary Chair

Berthold K.P. Horn (MIT, USA)

General Chairs

Roy E. Welsch ((MIT, USA)

Kazuya Takeda (Nagoya Univ., Japan)

Ahmad Hussein (IAV GmbH , Germany)

Ozaki Nobuyuki (Nagoya Univ., Japan)

Ji Li (Tsinghua Univ., China)

Lionel C. Kimerling (MIT, USA)

Shuguang Zhang (MIT, USA)

Technology Chairs

Yajun Fang (UVS, USA)

Xiaoman Duan (UVS, USA)

Yasha Yi (UWM, USA)

Conference Chairs

Lin Zhang (Beihang Univ., China)

Clinton J. Andrews (Rutgers Univ., USA)

Juejun Hu (MIT, USA)

Francisco B. Barnes (Tenure&Ecology LLC, Spain)

Executive Conference Chairs

Lin Zhang (Univ. at Buffalo, USA)

Longfei Zhou (UNF, USA)

Viksit Kumar (Novateur, USA)

Local Chair

Wei Wang (UVS, USA)

IEEE UV2026 General Committee

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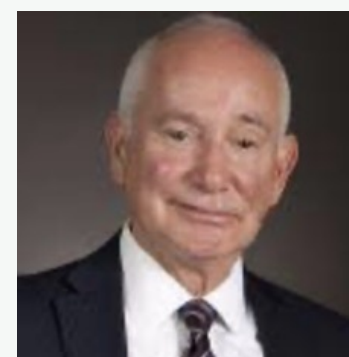
Ahmed Hussein
(IEEE ITSS , USA)



Ozaki Nobuyuki
(Nagoya Univ., Japan)



Ji Li
(Tsingda, China)



Lionel C. Kimerling
(MIT, USA)



Shuguang Zhang
(MIT, USA)

IEEE UV2026 International Committee

International Chairs



Yajun Fang
(UVS, USA)



Xiaoman Duan
(UVS, USA)



Yahsa Yi
(UMich, USA)

IEEE UV2026 Conference Committee

Conference Chairs



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(Beihang Univ., China)



Clinton Andrews
(Rutgers Univ., USA)



Juejun Hu
(MIT, USA)



Francisco Bozzano Barnes
(Tenure&Ecology LLC, Spain)

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Executive Conference Chairs



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(MIT, USA)*



*Longfei Zhou
(Gannon Univ., USA)*



*Viksit Kumar
(Novateur, USA)*

Local Chairs



*Wei Wang
(Robostreet, USA)*

IEEE UV2026 Program Committee

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Program Chairs



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(Ningbo Univ., China)



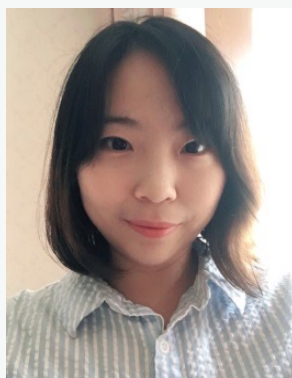
Zili Li
(MIT, USA)



Hao Sheng
(Beihang Univ., China)



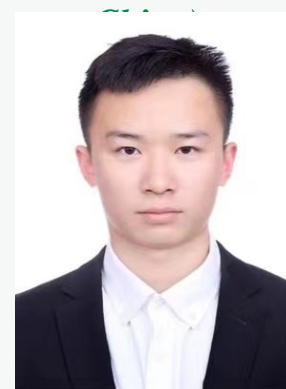
Hirofumi Aoki
(Nagoya Univ., Japan)



Wenya Du
(MIT, USA)



Yi Tao
(UVS, USA)



Juntao Jiang
(UVS, USA)



Zhenyao Liu
(UPENN., USA)

IEEE UV2026 Program Committee

Publication Chairs



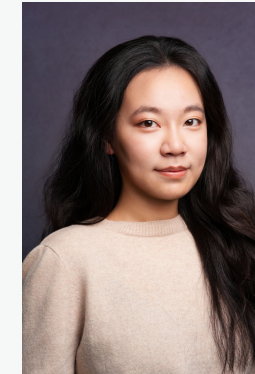
Hao Yuan
(Stevens Inst. of
Tech., USA)



Hongyan Cui
(MIT/BUPT,
USA/China)

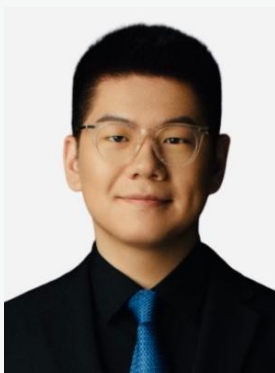


Lifeng Zhang
(UVS, USA)



Ge Cheng
(UVS, USA)

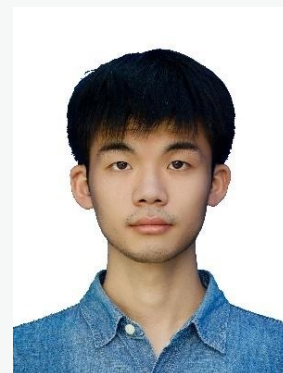
IT Chairs



Jieren Kou
(UVS, USA)



Hanxia Li
(UVS, USA)



Zhenqian Huang
(UVS, USA)



Chuqiao Gu
(Google, USA)

IEEE UV2026 Program Committee

Administrative Chairs

- ❖ Jason Chang (UVS, USA)
- ❖ Ryan VanBusum (UVS, USA)
- ❖ Yilin Wang (UVS, USA)

Education Chairs

- ❖ Fan Chen (Harvard Univ., USA)
- ❖ Jenny Wan (Sage Hill Consulting, USA)
- ❖ Bali Nelson (Nelson ESL Consulting, USA)

Public Relation Chairs

- ❖ Yohji Suhara (Community Technology Institute, Japan)
- ❖ Javier Sanchez Medina (EU/ULPGC, Spain)
- ❖ Toshimitsu Hamada (Tsukuba Gakuin Univ., Japan)
- ❖ Junyi Yu (Columbia Univ., USA)
- ❖ Mingyuan Hu (UVS-Nanjing, China)
- ❖ Weimin Hong (UVS, USA)
- ❖ Yulan Yang (SQIAST, China)

City Forum Chairs

- ❖ Huibo Liu (Fortune Talent LLC, China)
- ❖ Heng Ding (Joy Wisdom, China)
- ❖ Qinghe Kong (Jiyuan, China)

Media Chairs

- ❖ Yanyan Li (Sichuan Univ., China)
- ❖ Ronghua Chen (Oriental Foco Films, Singapore)
- ❖ Lan Huo (UVS, USA)
- ❖ Benjamin Cheung (Architectural Illusions, USA)

Financial Chairs

- ❖ Zeyao Wang (Duke Univ., USA)
- ❖ Sophia Wei (UVS, USA)

Panel Discussion Chairs

- ❖ Gene Fry (Global Warming Consultant, USA)

IEEE UV2026 Dates and Venues

Dates: October 19 - 22 , 2026

**Following the Successful IEEE UV2022,
IEEE UV2026 will be Held
Virtually & Locally**

Email : uv.conf.committee@universal-village.org

Website : <https://www.UniversalVillage.org>

UV Program

Pt. I

Keynote Day

Technical Sessions

Special Sessions

Pt. II

UV Forums

1. UV City Forum
2. UV Student Forum
3. UV Art Forum

UV Competitions

1. UV Data Science Competition
2. UV Modeling Competition
3. UV Pitch Competition
4. UV K-12 Challenge

UV Workshop

UV Poster & Exhibition

Pt. III

UV Roadmap Discussion

IEEE UV2026 Basic Structure of Sessions

Keynote Day

- ❖ Organized by IEEE UV2026 program committee
- ❖ Opening remarks and invited keynote talks

Technical Sessions

- ❖ Organized by session chairs
- ❖ Distribute “Session Call-for-Papers” by session chairs
- ❖ Session chairs select oral talks and review submitted papers

Special Sessions

- ❖ Distribute “Call-for-Session-Proposals” by the program committee
- ❖ Special sessions organized by special session chairs
- ❖ Special session chairs select oral talks, panels, and organize paper submissions

Technical Sessions (A)

1. [Session 1] UV Vision, Development, and Evaluation

- ❖ [TS1-A] Vision for Universal Village
- ❖ [TS1-B] Development Status of Universal Village
- ❖ [TS1-C] UV Indices, UV Incentive Mechanisms, and Evaluation of UV Subsystems - Methods, Technologies, and Systems

2. [Session 2] UV Framework Design, Modeling, and System Integration

- ❖ [TS2-A] Systematic and Integrated Frameworks for UV Subsystems and Contributing Factors
- ❖ [TS2-B] Intelligent Modeling, Simulation, and System Analysis
- ❖ [TS2-C] Scalability, Sustainability, and Human-Centered Design in UV System Integration
- ❖ [Special Session] [TS2-SP1] Intersection of Energy, Power Science and Smart Cities

3. [Session 3] Mobility, Planning, Management, and Infrastructure Innovations

- ❖ [TS3-A] Intelligent Transportation, Urban Planning, and Smart City Infrastructure
- ❖ [TS3-B] Intelligent Vehicles, Mobility Support for Vulnerable Groups
- ❖ [TS3-C] Smart Response Systems for City Emergencies
- ❖ [TS3-D] Information Flow, Communication, Networks, and Security
- ❖ [TS3-E] UV Underground: Underground Infrastructure & Future Urbanization Needs
- ❖ [Special Session] [TS3-SP1] Built Environment, Travel Behavior, and Data-Driven Mobility Planning
- ❖ [Special Session] [TS3-SP2] Sustainable Built Environments: Algorithmic Modeling and Generative AI

Technical Sessions (B)

4. **[Session 4] Energy and Material Innovations for Efficiency, Safety, and Sustainability**
 - ❖ [TS4-A] Renewable Energy and Smart Energy Management
 - ❖ [TS4-B] Smart Materials and Devices
 - ❖ [Special Session] [TS4-SP1] Semiconductor Chip Manufacturing in the AI Era: Enabling 3D Optoelectronics & Next-Generation Integration
5. **[Session 5] Manufacturing Innovations for Efficiency, Safety, and Sustainability**
 - ❖ [TS5] AI-Enabled Manufacturing
6. **[Session 6] Agriculture Innovations for Efficiency, Safety, and Sustainability**
 - ❖ [TS6] AI-Driven Smart Agriculture: Bridging Tech Innovation & Sustainable Growth for Rural Vitalization
7. **[Session 7] Environmental Innovations for Harmony between Humans and Nature**
 - ❖ [TS7-A] Smart Environmental Protection; Smart Ecological Systems
 - ❖ [TS7-B] Mobility-enabled Material Cycles; the Circular Economy; Trash and Scrap Collection, Processing, Reuse, and Recycling
8. **[Session 8] Lifestyle Style Innovations for Happiness**
 - ❖ [TS8-A] Smart Homes and Community, Virtual Living
 - ❖ [TS8-B] Mobility, Connectivity, and Innovative Lifestyles
 - ❖ [Special Session] [TS8-SP1] Integration of Sports Science and Medical Research: Interdisciplinary Innovation in Sports, Traditional Chinese Medicine, and Western Medicine

Technical Sessions (C)

9. [Session 9] *Healthcare Innovations for Safety and Well-Being*

- ❖ [TS9-A] AI-assisted Healthcare Monitoring
- ❖ [TS9-B] Smart Medicine and Smart Healthcare
- ❖ [TS9-C] Public Health, Epidemic Prevention and Control
- ❖ [TS9-D] The Fusion of Traditional Chinese Medicine and Modern Science
- ❖ [TS9-E] AI-Driven Sports Training, Exercise, and Fitness
- ❖ [Special Session] [TS9-SP1] AI-Powered IoMT Telehealth Innovation

10. [Session 10] *Social and Cultural Revolution for Diversity, Inclusiveness, Fairness, and Preservation of Cultural Heritage*

- ❖ [TS10-A] Urbanization, Social Governance, and Smart Communities
- ❖ [TS10-B] Smart Government and Social Services
- ❖ [TS10-C] Integrated Solutions for Smart Humanity
- ❖ [TS10-D] Smart Design and Design Ethics
- ❖ [TS10-E] Cultural Heritage Preservation and Creative Industries
- ❖ [TS10-F] Equitable AI for the Underrepresented and Vulnerable Groups

Technical Sessions (D)

11. [Session 11] Algorithms, Benchmark Development, and Data Management

- ❖ [TS11-A] Responsible and Ethical Data Management and Processing
- ❖ [TS11-B] Learning Algorithm and Model Development, Analysis, and Interpretability
- ❖ [TS11-C] Benchmark Development, Evaluation, and Long-Term Reproducibility

12. [Session 12] Special Sessions - New Proposed Sessions

- ❖ [SP12-A] AI and Finance
- ❖ [SP12-B] Reconstructing Human-Centered Family Education in the AI Era: Learning, Parenting, and Relational Development
- ❖ [SP12-C] Shanwei Forum: Quality-Driven Development of New Energy Vehicles through Sustainable Smart Manufacturing

Roles & Responsibilities for UV2026 Session Chairs

Timetable for Session Chairs

Date	Responsibilities / Tasks
August 1	Finalize speaker and panelist profiles, including bios, affiliations, profile photos, talk titles, and abstracts.
September 1	Conclude the peer-review process for manuscripts submitted by the official August 1 deadline. Issue official acceptance notifications to authors.
September 5	Submit proposed session agendas to the Program Chairs, including preferred scheduling, speaker order, presentation durations, and total session length.
September 15	Conclude the review process for Program Chair-approved late submissions, if applicable. (Round 1) Review final program arrangements and update session agendas, speaker titles, and abstracts.
September 30	Complete any remaining review updates for approved late submissions, if applicable. (Round 2) Review and update the official Conference Program Book materials.
October 7	Finalize remaining session updates and submit session documentation to the Program Chairs. (Round 3) Compile and submit session proceedings materials, subject to Program Chairs' confirmation.
October 10–11	Conduct technical connectivity tests with session speakers and collect backup presentation videos.
October 17–20	[Official IEEE UV2026] Execute the live session, manage Q&A, and ensure adherence to the published schedule.
November 1	Collect post-event revised files and final archive materials from invited speakers and general authors, if required.
November 8	Submit updated session documentation and folders to the Program Chairs. Provide any necessary technical clarifications for the final archive.

Timetable for Program Chairs

Date	Responsibilities / Tasks
August 1	Review and approve the roster of invited speakers to ensure technical alignment with UV2026 themes.
September 1	Distribute technical instructions for preparing camera-ready manuscripts. Issue formal registration reminders to all prospective attendees and authors.
September 5	Collect and synthesize session program proposals from Session Chairs to begin building the master schedule.
September 15	Consolidate session program updates and synthesize the master agenda. Finalize scheduling and time slots with all Session Chairs.
September 25	Commence formal compilation of the official Conference Program Book.
October 1	Update and digitally publish the Conference Program Book on the official website. Compile online meeting instructions and notify all presenters and attendees.
October 10	Update the Conference Program Book on the official website. Compile preliminary Conference Proceedings and generate secure download access for registered participants.
October 10–11	Conduct technical connectivity tests with keynote speakers and collect backup video presentations.
October 17–20	[Official IEEE UV2026] Coordinate the official Sessions and Keynote Program of IEEE UV2026.
November 1	Audit collected manuscripts for IEEE formatting and copyright compliance.
November 17	Incorporate post-event updates into the definitive Program Book and Conference Proceedings. Submit the final proceedings package to IEEE for possible archival and inclusion in IEEE Xplore, subject to IEEE scope, formatting, copyright, and quality requirements.

Roles & Responsibilities of Session Chairs

【1】 Session Organization

- ☐ **Prepare** and distribute the session-specific “Call for Papers” poster. *
- ☐ **Prepare and distribute** the “Session Committee & Invited Speakers” poster. *
- ☐ **Invite** authors, session committee members, speakers, and reviewers.
- ☐ **Collect and submit** speaker bios, profile photos, talk titles, and abstracts to the Program Chairs.
- ☐ **Organize** paper reviews through the online review system. *
- ☐ **Submit** the proposed session program and agenda to the Program Chairs.
- ☐ **Ensure** that authors follow IEEE paper formatting and copyright requirements for camera-ready papers. *
- ☐ **Remind** speakers and authors to register early to receive the early-bird rate.

** [To be completed together with Session Assistants]*

Roles & Responsibilities of Session Chairs

【2】 Hosting Preparation

- ❑ **Keep** Program Chairs updated on session progress and any program changes.
- ❑ **Notify** speakers about their assigned time slots and confirm their availability. *
- ❑ **Ensure** stable connections for remote presenters and stable performance of on-site audio/video systems. *
- ❑ **Collect** backup presentation videos. *
- ❑ **Prepare** roundtable discussion topics. *

** [To be completed together with Session Assistants]*

Roles & Responsibilities of Session Chairs

【3】 Online/Hybrid Session Hosting

- ☐ **Announce** the beginning of the session and introduce the session topic.
- ☐ **Provide** a brief summary of the session.
- ☐ **Introduce** each speaker and presentation title, and host the Q&A session.
- ☐ **Prepare** constructive questions for authors in case there are no questions from the audience.
- ☐ **Maintain** the published schedule and be prepared for unexpected situations.
- ☐ If a presenter **does not show up**, record the no-show in the Session Report Form and do not move to the next scheduled presentation.
- ☐ **Use backup activities** if needed, such as extended Q&A, prepared discussion topics, or brief session reflections.
- ☐ **Provide** closing remarks, thank speakers and attendees, and submit the session report.

Roles & Responsibilities for UV2026 Session Assistants

Roles & Responsibilities of Session Assistants

【1】 Session Organization

- ❑ **Design and update** session posters using the UV2026 poster template.
- ❑ **Update** session posters with bios and profile photos of session committee members and invited speakers.
- ❑ **Monitor** paper submission status and **report** updates to Session Chairs.
 - ❖ **Report** duplicate submissions and technical issues to the IT Team.
- ❑ **Provide** technical support for the online review system for Session Chairs and reviewers.
- ❑ **Assist** authors in preparing camera-ready papers that meet IEEE formatting and copyright requirements.
- ❑ **Assist** with UV2026 registration and payment questions for session speakers, and presenters, when needed.
- ❑ **Collect** backup presentation videos and related session materials.

【2】 Hosting Preparation Support

- ❑ **Assist** Session Chairs in notifying speakers about assigned time slots and confirming availability.
- ❑ **Test and support** stable connections for remote presenters and stable performance of on-site audio/video systems.
- ❑ **Prepare speaker materials** for Session Chairs in presentation order, including bios, profile photos, presentation files, and backup videos.
- ❑ **Provide technical support** to Session Chairs and speakers before and during the session.
- ❑ **Keep backup files** organized and accessible in case of technical issues during the live session.

Registration Fee

IEEE UV2026	Conference Registration Fee for Authors and Presenters		Paper Publication Fee for Authors (<i>N = Number of Accepted Papers</i>)	Conference Registration Fee for Attendee Only	
	Early Bird Rate [Before Sept.1]	Standard Rate [After Sept.1]		Whole Session Pass	One Day Pass
Non-IEEE members	\$695	\$795	+ \$150 * N	\$400	\$200
IEEE/UV members	\$595	\$695	+ \$150 * N	\$400	\$200
IEEE ITSS members	\$545	\$645	+ \$150 * N	\$400	\$200
Non-IEEE Student members	\$395	\$445	+ \$150 * N	\$275	\$100
IEEE/UV Student members	\$325	\$375	+ \$150 * N	\$275	\$100
K-12	\$195	\$245	+ \$150 * N	\$175	\$100

*All fees are in USD and subject to confirmation on the official UV2026 registration page.

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INTERNATIONAL CONFERENCE ON UNIVERSAL VILLAGE



Thank You!

*Let's Work Together to Make IEEE UV2026
Successful!
Another UV Milestone!*